Pratik Nanaso Raut

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PROFESSIONAL SUMMARY

Computer Science candidate with strong foundation in C++, Data Structures, OS Concepts — systems development. Hands-on experience building concurrent servers using POSIX threads at Gentrack.

TECHNICAL SKILLS

Languages	Java, C, C++, SQL, JavaScript, Python (basics)
Backend	Spring Boot, REST APIs, Spring Security, Hibernate, JWT, JDBC
Databases	MySQL, MongoDB
Tools	Git, GitHub, Maven, Postman, IntelliJ IDEA
Concepts	Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), SDLC, Design Patterns, Problem Solving, Time Complexity Analysis, REST API Design, Database Optimization, RBAC, JWT Authentication, Microservice Fundamentals, API Security, Multi-Tenant Architecture, Multithreading

PROFESSIONAL EXPERIENCE

Software Developer Intern *Feb 2026 -- Ongoing*

SetTribe IT Solutions | Pune

- Developed and enhanced ERP backend modules using Java, Spring Boot, Hibernate, and REST APIs for employee and operational workflows.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

ACADEMIC PROJECTS

Customised Virtual File System (CVFS) -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations using C++ and inode-based metadata management.
- Achieved O(1) file lookup by designing hash table-based inode indexing.
- Implemented in-memory file allocation table managing file permissions and directory traversal.

ProcMemInspector -- Linux Process Memory Layout Analyzer *C, Linux Internals, /proc Filesystem*

- Developed /proc filesystem parser in C to display virtual memory layout.
- Implemented process statistics monitoring by reading /proc/[pid]/maps.
- Analyzed virtual address space segmentation demonstrating memory isolation.

Concurrent FTP Server -- Multi-Client File Transfer Protocol Server *C, POSIX Threads, TCP Sockets, Multithreading*

- Engineered concurrent client handling using POSIX threads over TCP sockets.
- Implemented thread-safe resource management using mutex and semaphore primitives.
- Designed application-layer protocol handling authentication and file transfer commands.
- Strong foundation in C++ OOP, STL containers, templates, and dynamic memory management.
- Proficient in POSIX system calls, concurrent programming, and Linux kernel interface design.
- Hands-on experience implementing OS-level components: file systems, network servers, and memory analyzers.
- Practical exposure to network programming, socket communication, and concurrent server development.

EDUCATION

Master of Computer Applications (MCA) *2026 | Pursuing*

Dnyansagar Institute of Management and Research, Pune (SPPU) 7.24 CGPA

Bachelor of Computer Applications (BCA) *2024*

C.T. Bora College, Shirur (SPPU) 73.34%

Higher Secondary Certificate (HSC) *2021*

Maharashtra State Board 75%